

UNIVERSITY OF THE ARMED FORCES - ESPE

Department of Exact Sciences

Object - Oriented Programming

NRC: 368663

WorkShop16-Unit Testing

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WORKSHOP 16 – UNIT TESTING

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Subject: Object-Oriented Programming
Topic: Unit Testing
Development Tools: Apache NetBeans, Visual Studio Code, Java, and Python

1. Euler's Buckling Formula

Euler's Buckling Formula is used to calculate the critical load at which a column may buckle under compression.

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

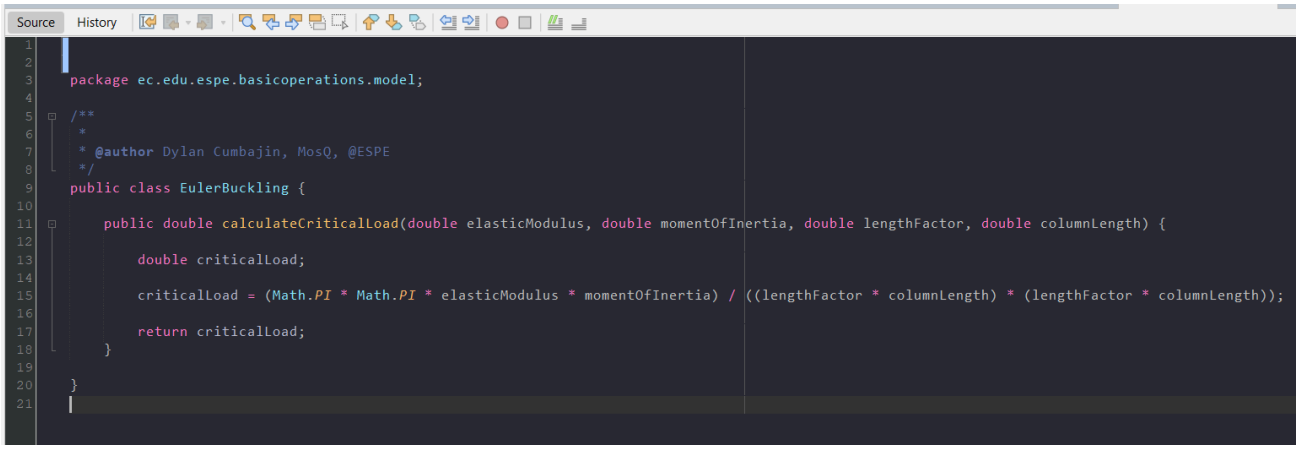
Where:

- **Pcr:** Critical load (N).
- **E:** Modulus of elasticity (Pa).
- **I:** Moment of inertia (m^4).
- **K:** Effective length factor.
- **L:** Column length (m).

2. Unit Testing in Java

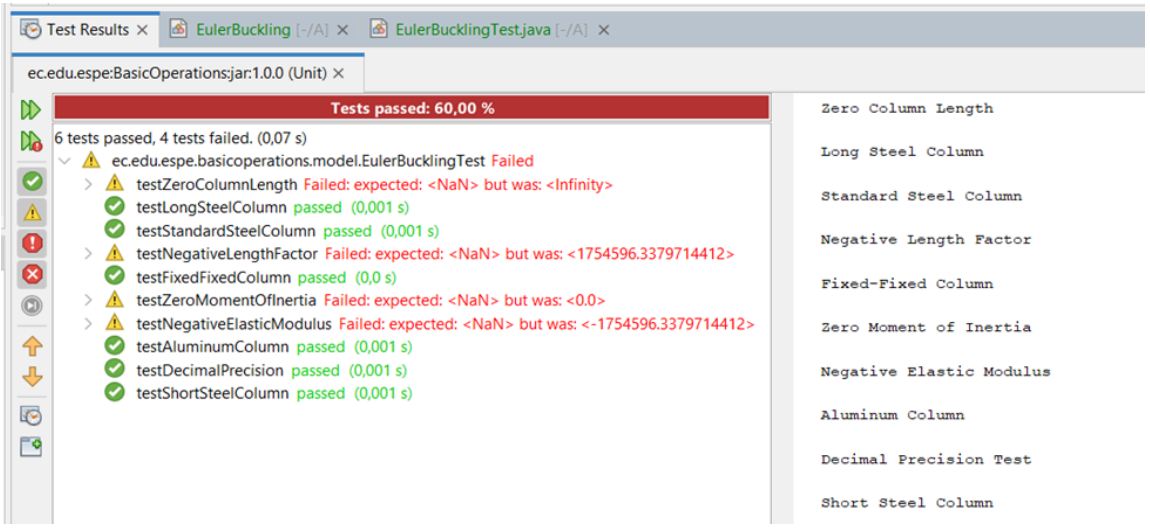
2.1. Test Cases

Figure 2. Ten Test Cases for Euler's Buckling Formula



2.2. Test Results

Figure 3. JUnit Test Results in Apache NetBeans



3. Unit Testing in Python

3.1. Test Cases

Figure 4. Python Unit Test Cases Using unittest

```python
import math


class EulerBuckling:

    def calculateCriticalLoad(self, elasticModulus, momentOfInertia, lengthFactor, columnLength):

        criticalLoad = (math.pi * math.pi * elasticModulus * momentOfInertia) / ((lengthFactor * columnLength) * (lengthFactor * columnLength))

        return criticalLoad
```

3.2. Test Results

Figure 5. Python Unit Testing Results in Visual Studio Code

```
UnitTesting\ws16UnitTestingPython"
dy456 ~\Desktop\ESPE2609-OOPSW36863\ws\cumbajin\U1\ws16UnitT❯ py -m unittest discover -s tests -p "test_*.py" -v
test01ShortSteelColumn (test_euler_buckling.TestEulerBuckling.test01ShortSteelColumn) ... ok
test02SteelColumn (test_euler_buckling.TestEulerBuckling.test02SteelColumn) ... ok
test03AluminumColumn (test_euler_buckling.TestEulerBuckling.test03AluminumColumn) ... ok
test04ZeroColumnLength (test_euler_buckling.TestEulerBuckling.test04ZeroColumnLength) ... ERROR
test05NegativeElasticModulus (test_euler_buckling.TestEulerBuckling.test05NegativeElasticModulus) ... FAIL
test06FixedFixedColumn (test_euler_buckling.TestEulerBuckling.test06FixedFixedColumn) ... ok
test07LongSteelColumn (test_euler_buckling.TestEulerBuckling.test07LongSteelColumn) ... ok
test08ZeroMomentOfInertia (test_euler_buckling.TestEulerBuckling.test08ZeroMomentOfInertia) ... FAIL
test09NegativeLengthFactor (test_euler_buckling.TestEulerBuckling.test09NegativeLengthFactor) ... FAIL
test10DecimalPrecision (test_euler_buckling.TestEulerBuckling.test10DecimalPrecision) ... ok

======================================================================
ERROR: test04ZeroColumnLength (test_euler_buckling.TestEulerBuckling.test04ZeroColumnLength)
----------------------------------------------------------------------
Traceback (most recent call last):
  File "C:\Users\dy456\Desktop\ESPE2609-OOPSW36863\ws\cumbajin\U1\ws16UnitTesting\ws16UnitTestingPython\tests\test_e
```

4. Unit Testing in TypeScript

4.1. Test Cases

Figure 6. TypeScript Unit Test Cases

```typescript

/**
 * @author Dylan Cumbajin, MosQ, @ESPE
 */

export class EulerBuckling {

    calculateCriticalLoad(elasticModulus: number, momentOfInertia: number, lengthFactor: number, columnLength: number): number {

        let criticalLoad: number;

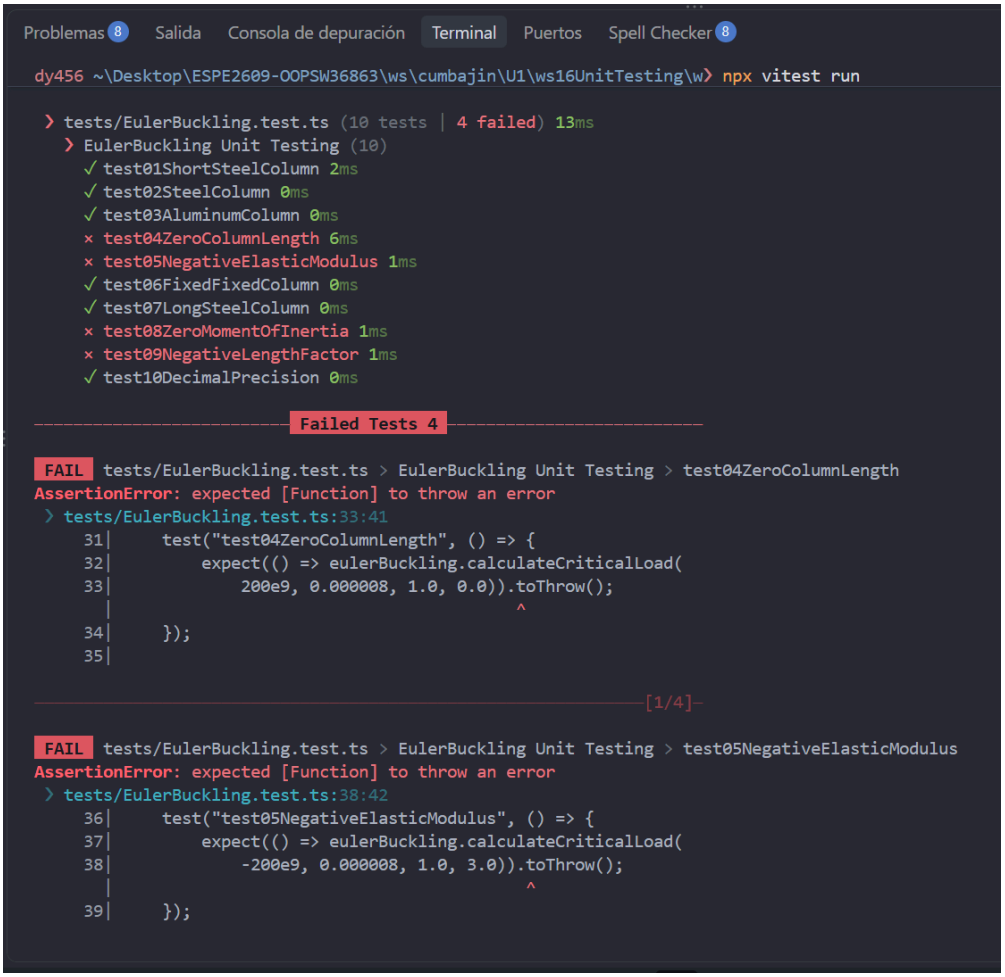
        criticalLoad = (Math.PI * Math.PI * elasticModulus * momentOfInertia) / ((lengthFactor * columnLength) * (lengthFactor * columnLength));

        return criticalLoad;
    }
}

```

4.2. Test Results

Figure 7. TypeScript Unit Testing Results



Test Case:

TEST CASE 01

Input Data		
Elastic Modulus (E)	=	200 GPa
Moment of Inertia (I)	=	0.000008 m^4
Length Factor (K)	=	1.0
Column Length (L)	=	3.0 m

TEST CASE 02

Input Data		
Elastic Modulus (E)	=	210 GPa
Moment of Inertia (I)	=	0.000005 m^4
Length Factor (K)	=	1.0
Column Length (L)	=	2.0 m

TEST CASE 03

Input Data		
Elastic Modulus (E)	=	70 GPa
Moment of Inertia (I)	=	0.000003 m^4
Length Factor (K)	=	1.0
Column Length (L)	=	1.5 m

TEST CASE 04

Input Data		
Elastic Modulus (E)	=	200 GPa
Moment of Inertia (I)	=	0.000008 m^4
Length Factor (K)	=	1.0
Column Length (L)	=	0.0 m

TEST CASE 05

Input Data		
Elastic Modulus (E)	=	-200 GPa
Moment of Inertia (I)	=	0.000008 m^4
Length Factor (K)	=	1.0
Column Length (L)	=	3.0 m

TEST CASE 06

Input Data		
Elastic Modulus (E)	=	200 GPa
Moment of Inertia (I)	=	0.000008 m^4
Length Factor (K)	=	0.5
Column Length (L)	=	3.0 m

TEST CASE 07

Input Data		
Elastic Modulus (E)	=	200 GPa
Moment of Inertia (I)	=	0.000008 m^4
Length Factor (K)	=	1.0
Column Length (L)	=	6.0 m

TEST CASE 08

Input Data		
Elastic Modulus (E)	=	200 GPa
Moment of Inertia (I)	=	0.0 m^4
Length Factor (K)	=	1.0
Column Length (L)	=	3.0 m

TEST CASE 09

Input Data		
Elastic Modulus (E)	=	200 GPa
Moment of Inertia (I)	=	0.000008 m^4
Length Factor (K)	=	-1.0
Column Length (L)	=	3.0 m

TEST CASE 10

Input Data		
Elastic Modulus (E)	=	205 GPa
Moment of Inertia (I)	=	0.0000075 m^4
Length Factor (K)	=	0.7
Column Length (L)	=	2.75 m